Project Panopticon

Intelligent Exam Proctoring: time-series ML that protects innocent students

EduGuard AI, Machine Learning Division | Intern: Devyani Nandkishor Vighe| Date:07/10/2026

1. Project Problem Statement

- Online proctoring tracks eye gaze, background audio and computer logs to detect cheating.
- The current system flags thousands of innocent students for sneezing, reading a question aloud or glancing at the keyboard, and universities are threatening to cancel contracts.
- Two data problems: video telemetry (every 1 second) and system events (only when something happens) are asynchronous, and dropped video leaves large gaps of missing values.
- Two modelling problems: a 1-second glance is not cheating, and only 77 of ~10,000 seconds (under 1%) are cheating, so accuracy is misleading.
- A false accusation can destroy an academic career, so the system must flag a student only when it is highly certain (90%+).

2. Project Solution and Tools

- Align: pd.merge_asof (backward, 1-second tolerance) attaches each system event to the nearest video second without smearing labels.
- Clean: forward-fill eye gaze, linearly interpolate audio, fill missing tab_switches / is_cheating with 0. The timeline is kept (732 dropped seconds recovered).
- Engineer: 10-second rolling mean of gaze and rolling max of audio, so only sustained behaviour counts.
- Model: RandomForestClassifier with class_weight='balanced', 80/20 stratified split, random_state=42.
- Decide: predict_proba with a custom 0.90 threshold instead of the default 0.50, plus a Precision-Recall curve.
- Tools: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Google Colab / Jupyter.

3. Project Screenshots: Data Alignment

```
 Running locally. Ensure CSV files are in the same directory.
Video telemetry: (10000, 3) | System events: (167, 3)
Missing values in video telemetry:
 timestamp          0
eye_gaze_angle    732
audio_db          732
dtype: int64
Events in source: 167 | Events aligned: 167

            timestamp  eye_gaze_angle   audio_db  tab_switches  is_cheating
0 2023-12-01 08:00:00        2.560857  36.553781           NaN          NaN
1 2023-12-01 08:00:01       10.177137  39.102627           NaN          NaN
2 2023-12-01 08:00:02        5.761714  31.228565           NaN          NaN
3 2023-12-01 08:00:03        9.787105  34.051405           NaN          NaN
4 2023-12-01 08:00:04        4.572665  35.245018           NaN          NaN
```

```
 Feature engineering complete. Current dataframe shape: (9991, 7)
Remaining NaN values: 0
Class balance:
 is_cheating
0.0    9914
1.0      77
Name: count, dtype: int64
```

```
 Model training complete.
Feature importances:
tab_switches          0.557
audio_db              0.222
eye_gaze_angle        0.138
gaze_rolling_10s      0.075
audio_rolling_10s     0.008
dtype: float64
```

Left: asynchronous merge, all 167 events aligned. Right: feature engineering (no NaNs left, 77 cheating rows) and model training.

3. Project Screenshots: Threshold Tuning

```
 DEFAULT THRESHOLD (0.50)
[[1978    6]
 [   1   14]]
              precision    recall  f1-score   support

           0       1.00      1.00      1.00      1984
           1       0.70      0.93      0.80        15

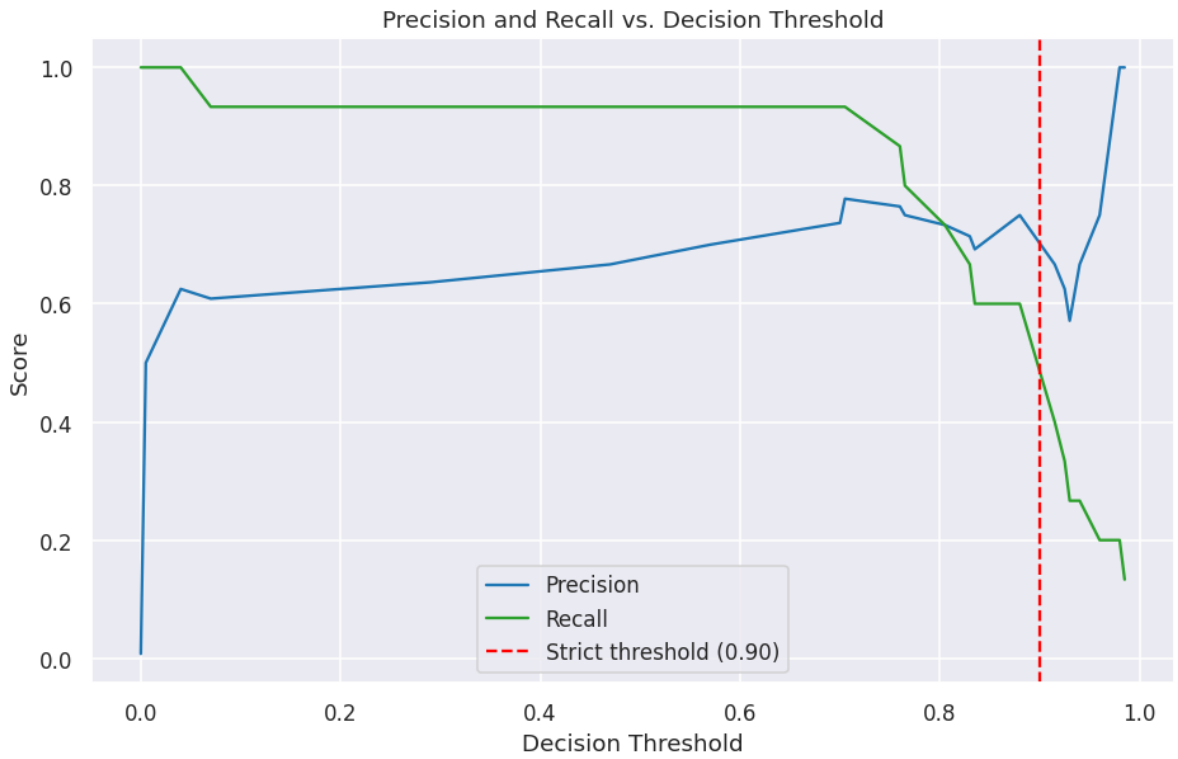
    accuracy                           1.00      1999
   macro avg       0.85      0.97      0.90      1999
weighted avg       1.00      1.00      1.00      1999
```

```
STRICT THRESHOLD (0.90)
[[1981    3]
 [   9    6]]
              precision    recall  f1-score   support

           0       1.00      1.00      1.00      1984
           1       0.67      0.40      0.50        15

    accuracy                           0.99      1999
   macro avg       0.83      0.70      0.75      1999
weighted avg       0.99      0.99      0.99      1999

False Positives: 6 (0.50)  ->  3 (0.90)
```



Raising the threshold from 0.50 to 0.90 cut test-set false positives from 6 to 3. Red dashed line marks 0.90.

3. Project Screenshots: Robustness Check

```
Threshold 0.50: flagged= 79 | TP= 68 | FP= 11 | precision=0.86 | recall=0.88
Threshold 0.90: flagged= 31 | TP= 29 | FP=  2 | precision=0.94 | recall=0.38
```

- Only ~15 cheaters fall in the test split, so a single split is noisy.
- 5-fold cross-validation over all 9,991 rows is a steadier estimate.

4. Project Report

- Result at 0.50 (cross-validated): precision 0.86, recall 0.88, 11 innocent students flagged.
- Result at 0.90 (cross-validated): precision 0.94, recall 0.38, only 2 innocent students flagged.
- Single test split at 0.90: 3 false positives, precision 0.67 on 9 flags. This is a small-sample effect, which is why cross-validation was added.
- Top features: tab_switches, audio_db, eye_gaze_angle, 10s gaze rolling mean.
- Trade-off: the strict threshold misses some cheaters on purpose. Flags should go to a human reviewer, not trigger automatic penalties.
- Recommendation: GO, as a human-in-the-loop review tool at the 0.90 threshold.

5. Video Explanation

- Link to 2-5 minute walkthrough video: https://drive.google.com/file/d/1EpJIcCkBan0Mhb_dhe45v_EpdhYsvOIE/view?usp=sharing
- Covers: the problem, the four traps, a live run of the notebook, and the threshold comparison.